

VEDANT GUPTA

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EDUCATION

Delhi Technological University

B.Tech in Mathematics & Computing Engineering

New Delhi, India

2023 – 2027

CGPA: 9.083/10

- **Relevant Coursework:** Object-Oriented Programming, Computer Networks, Operating Systems, Database Management Systems, and Data Structures & Algorithms.

Aryan International School

Class 12th, CBSE: 90.0%

New Delhi, India

2023

Christ Church Boys' Senior Secondary School

Class 10th, CBSE: 95.5%

Jabalpur, India

2021

EXPERIENCE

Center for Railway Information Systems (CRIS)

AI Full Stack Developer Intern

Delhi, India

June 2026 – July 2026

- Engineered an agentless Linux telemetry platform (**Node.js**, **Express**, **Socket.io**) via async SSH2 polling, streaming CPU/RAM/Disk metrics at **sub-50ms latency** for real-time system-health visibility.
- Implemented a multi-agent **RAG** (Retrieval-Augmented Generation) AI diagnostic pipeline (**LangGraph**, **LangChain**, **Qdrant** vector database), cutting troubleshooting from 15+ min to **sub-800ms**.
- Formulated a predictive maintenance engine using **Linear Regression** and **Z-score modeling**, forecasting disk partition exhaustion 7 days out to prevent server downtime.
- Collaborated with infrastructure engineers to integrate Linux telemetry into automated incident triage pipelines, deploying **node-cron** for real-time **Slack Block Kit** and SMTP alerts to reduce manual response time by **95%** (saving 10+ hrs/month).

PROJECTS

Botlock — Decentralized AI Agent Paywall & Micropayment System

[GitLab](#) | [Live](#)

Tech Stack: Node.js, Express.js, Solana Web3.js, Supabase, Next.js, Docker, Railway, Vercel

- Designed a content gating gateway protocol (**HTTP 402**, **HMAC-SHA256** challenges), verifying on-chain USDC via **Solana RPC** in **400ms** at **under \$0.001/tx**.
- Published two **npm packages** enabling **2-line SDK integration**, letting publishers adopt payment verification within minutes of installation.
- Constructed a multi-signal bot detection pipeline with $O(1)$ **bitwise CIDR matching** and a **2,000-entry LRU** reverse DNS queue, achieving an **under 0.1% false-positive rate** for humans.

NexusAI — Distributed Multi-Agent AI Platform with Microservices Orchestration

[GitLab](#) | [Live](#)

Tech Stack: Node.js, Express, React 19, Python, LangChain, LangGraph, Redis, MongoDB, AWS S3, Docker

- Architected **4 microservices** behind a **Redis** reverse-proxy API Gateway, eliminating credential propagation, and built a **Razorpay** billing service achieving **zero stale-state reads** via atomic plan sync.
- Assembled an **8-agent, multi-agent LangGraph** orchestration engine with a prompt-engineered two-phase supervisor router for deterministic file uploads and semantic classification, achieving **sub-200ms intent routing**.
- Secured system with **Redis atomic rate limiting** (3 to 20 req/min), tiered **idempotent** credit deduction, and a **20-message sliding-window** conversation memory (24h TTL).

FETCHIT — Cross-Platform Open Source Command Line Interface Video Downloader

[GitLab](#) | [Live](#)

Tech Stack: TypeScript, React 19, Ink, Node.js, Next.js, Tailwind, CI/CD, yt-dlp, FFmpeg, npm

- Spearheaded the launch of FETCHIT on **npm**, an **open-source** media CLI/TUI, enabling media downloading from **more than 2,000 websites**.
- Built a two-tier metadata probing engine and worker-pool batch queue (**–concurrency**), reducing playlist discovery latency **85%** (15s+ to **sub-800ms**).
- Optimized deterministic format selection by pinning **yt-dlp format IDs** to probed manifests, eliminating a **46x size discrepancy bug** (144p label delivering 232 MB 4K video).

TECHNICAL SKILLS

Languages: C/C++, JavaScript, TypeScript, Python, SQL, HTML, CSS

AI/ML & GenAI: RAG (Retrieval-Augmented Generation), Prompt Engineering, Multi-Agent Systems, Agent Orchestration, Loop Engineering, Vector Embeddings & Semantic Search, LangChain, LangGraph, LLM APIs

Frameworks & Libraries: React.js, Node.js, Express.js, Next.js, REST API Design, Socket.io, Framer Motion

Databases: PostgreSQL, MongoDB, Redis, Vector Databases (Qdrant), Supabase

Cloud & DevOps: AWS S3, Docker, Railway, Vercel, GitLab CI/CD

Tools & Platforms: Git, npm, Postman, Solana Web3.js, BullMQ, Linux

HONORS & AWARDS

- Solved **600+ Data Structures & Algorithms (DSA)** problems across multiple competitive programming platforms.
- Attracted **150+ GitHub stars** and **3,000+ npm downloads** for open-source project FETCHIT.
- Attained a perfect **10.0 SGPA** and secured **University Rank 1** in the 5th semester.
- Ranked in top **1.9% of over 1.1 million candidates** in JEE Mains 2023 and qualified prestigious **NTSE Stage-1** in 2021.